

Software Product

Project Name: **signify**

A real-time AI-powered overlay that translates system audio into 3D Sign Language animation, making any media accessible without content creator intervention.

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Definitions & Acronyms

STT (Speech-to-Text): The process of converting spoken audio signals into written text.

NLP (Natural Language Processing): Computational techniques used to analyze and manipulate human language.

Glossing: The linguistic notation used to transcribe Sign Language converting "I am going to the store" to the gloss structure "STORE ME GO".

IK (Inverse Kinematics): A mathematical process in 3D animation used to calculate the variable joint parameters (angles) needed to place the end of a kinematic chain (e.g., a hand) at a given position.

HDRP (High Definition Render Pipeline): A Unity rendering engine focused on high-fidelity, photorealistic graphics.

Introduction

Problem Statement & The Gap

Digital accessibility for the Deaf and Hard-of-Hearing community currently relies heavily on Captions. However, for many in the Deaf and Hard-of-Hearing community, Sign Language is their primary language, while written text is a secondary language that requires higher cognitive load and often lacks emotional intonation.

There is currently no system that functions as a universal OS-level "Overlay" capable of interpreting audio from any source (Zoom, Netflix, System Alerts, Local Video Files) in real-time.

The Solution & Added Value

Signify is a desktop application that acts as a "transparent layer" over the operating system. It captures system audio, translates it, and renders a 3D Avatar performing the signs in real-time.

- Immediate Value: The user does not need to wait for a content creator to upload a signed version of a video.
- Future Vision: The architecture is designed to be portable. While currently a PC application, the core logic is containerized, allowing future integration into Smart TVs as a native accessibility app.

Market Analysis ("Has this been done before?")

While text-to-sign dictionaries exist, the Real-Time Overlay Workflow is the primary innovation here. Most competitors focus on educational tools (teaching sign language) rather than live interpretation utilities. Signify bridges the gap between static translation and live media consumption.

Specification

User Personas & Scenarios

Scenario A: The Academic (Lecture Mode)

User: A Deaf and Hard-of-Hearing student attending a Zoom lecture where the professor does not have a sign language interpreter.

Flow: The student launches Signify. The Avatar floats in the bottom-right corner (Always-on-top). As the professor speaks, the Avatar translates the concepts into signs instantly.

Scenario B: Entertainment (Casual Mode)

User: A user watching a movie on a local media player (VLC) or streaming service.

Flow: The user connects Signify to the "System Audio" output. The Avatar provides signing alongside the movie subtitles.

High-Level Workflow

The data flows sequentially through five distinct stages to ensure low latency and high accuracy:

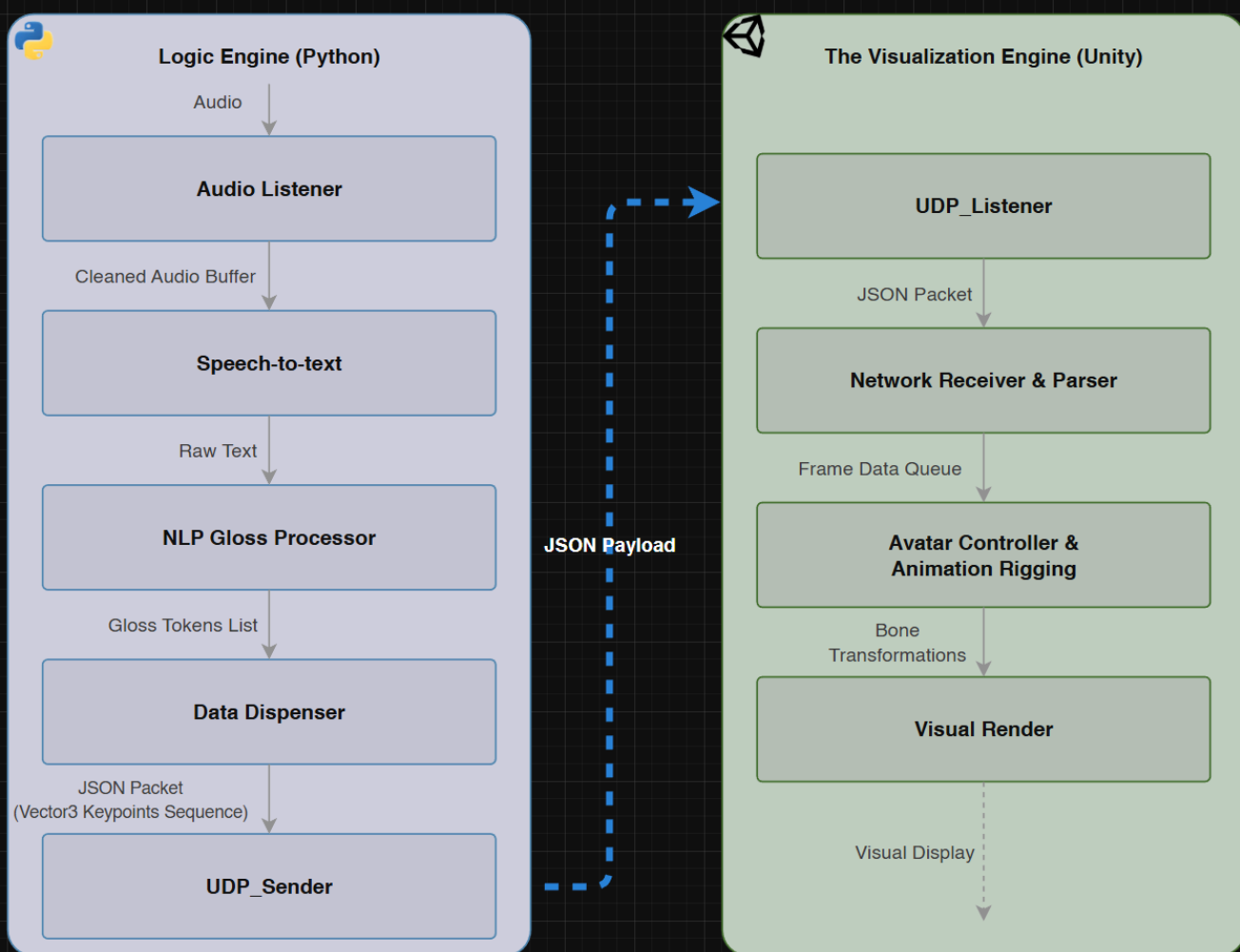
1. **Audio Capture:** The system hooks into the WASAPI (Windows Audio Session API) loopback to capture "what the user hears."
2. **Transcription (STT):** Audio buffers are sent to the Cloud (Google STT) to convert waves to raw strings.
3. **Linguistic Processing (The "Gloss" Layer):** Raw English is syntactically rearranged into Sign Language grammar (Subject-Object-Verb, removal of stop words).
4. **Kinematic Retrieval:** The system queries the local database for the skeletal animation data matching the Glosses.
5. **Rendering:** The Unity engine receives skeletal vectors and applies them to the Avatar using IK for smooth transitioning.

Design (Architecture & Components)

System Architecture

The system utilizes a Client-Side Micro-Modular Architecture. We decouple the logic (Python) from the presentation (Unity/C#) to allow for easier debugging and potential replacement of the rendering engine in the future.

Component Diagram:



Module A: The Logic Engine (Python Environment)

Component 1: Audio Listener

- Input: System Audio Stream (Bytes) / Microphone Input
- Logic: Noise cancellation, Silence detection (VAD - Voice Activity Detection).
- Output: Cleaned Audio Buffer (.wav chunk)

Component 2: STT Client (Cloud Connector)

- Input: Cleaned Audio Buffer
- Logic: API calls to Google Speech-to-Text (Streaming mode).
- Output: Raw String (e.g., "I am going home")

Component 3: NLP Gloss Processor

- Input: Raw String
- Logic: Stop-word removal, Reordering to Gloss syntax.
- Output: Gloss Token List (e.g., ["ME", "GO", "HOME"])

Component 4: Data Dispenser

- Input: Gloss Token List
- Logic: Search local JSON dataset. If a word is missing, fallback to "Finger Spelling" mode.
- Output: JSON Packet (Vector3 Keypoints Sequence)

Module B: The Visualization Engine (Unity)

Component 5: Network Receiver

- Input: JSON Packet (via UDP Socket)
- Logic: Deserialization of data into C# Structs.
- Output: Frame Data Queue

Component 6: Avatar Controller & Rigging

- Input: Frame Data Queue
- Logic: Smoothing: Linear interpolation between Frame A and Frame B to prevent jitter.
- Output: Visual Render (Pixels on Screen)

Data Design & Storage

Storage Format: Flat-file system.

Structure: `/assets/dictionary/{gloss_name}.json`

Content: Each JSON contains the XYZ coordinates of 54 wrist/finger landmarks relative to the root bone.

Technology Stack

Core Logic: Python

Rendering: Unity 2024 (C#)

Cloud Services: Google Cloud Platform (Speech-to-Text)

Libraries: `speech_recognition`, `spacy` (for NLP), `mediapipe` (for dataset generation), `Newtonsoft.JSON`.

MVP Definition (Proof of Concept)

Vocabulary: Top 100 most common daily words (Hello, Yes, No, Eat, Drink, Home, etc.).

Latency: Total processing time under 2 seconds.

Stability: The Avatar successfully transitions from "Idle" to "Signing" and back to "Idle" without crashing.

Open Questions & Future Risks

Handling Unknown Words: How do we handle names or slang not in the dictionary? (Proposed solution: Finger-spelling fallback).

Synchronization: Ensuring the speed of the animation matches the speed of speech (Time-stretching).